

Continuous-wavelet-transform-based automatic curve fitting method for laser-induced breakdown spectroscopy

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Received 29 May 2018; revised 11 August 2018; accepted 12 August 2018; posted 13 August 2018 (Doc. ID 332790); published 7 September 2018

In this work, an automatic curve fitting method based on a continuous-wavelet transform (CWT) is proposed to resolve overlapped peaks and to adaptively extract the major peaks in laser-induced breakdown spectroscopy (LIBS). From the local minimum of the second derivative of the LIBS spectrum calculated with CWT, the number of individual peaks is determined, and corresponding peak positions are estimated. The full width at half-maximums (FWHMs) of individual peaks are estimated from the separation of two maxima siding the minimum. A threshold is introduced to eliminate the small peaks and therefore reduce the number of fitting parameters and adaptively extract the major peaks with different spectral intensities. The Trust-Region algorithm is used for parameter optimization. The proposed method is used to analyze both simulated LIBS spectra and experimental overlapped peaks. Both simulated and experimental results show that the proposed method can resolve overlapped peaks even with a low separation degree, although the minimum resolvable separation degree depends on the FWHM ratio and strength ratio of individual peaks and the wavelet scale. In a LIBS calibration experiment of N₂/SF₆ gasses mixture, after resolving the overlapped peaks with the proposed method, better linear correlations between the concentration and intensity of F (with an adjusted R-squared value 0.9972), as well as between the concentration ratio and intensity ratio of nitrogen to fluorine (with adjusted R-squared values >0.98 and 0.99) are obtained. © 2018 Optical Society of America

OCIS codes: (300.6365) Spectroscopy, laser induced breakdown; (070.4790) Spectrum analysis; (140.0140) Lasers and laser optics.

<https://doi.org/10.1364/AO.57.007526>

1. INTRODUCTION

Laser-induced breakdown spectroscopy (LIBS) is widely used in many fields such as material analysis [1], environmental monitoring [2], biomedicine [3], archaeology [4], food safety [5], and so on. In principle, LIBS can analyze any matter, regardless of its physical state, be it solid [6], liquid [7] or gas [8], and it can detect all elements. However, the accuracy and reliability of the analysis are affected by peak overlapping, which commonly occurs in LIBS spectra due to various line-broadening mechanisms and the limitation of instrumental resolution. Besides, in many cases, the spectral analysis is performed manually and is therefore time-consuming, which makes online and real-time applications of LIBS somewhat difficult.

In the past, signal processing methods such as the iterative deconvolution method [9], Wiener filtering [10], and the curve fitting method [11–14] were widely used to resolve overlapped peaks in LIBS spectra. However, the effectiveness of iterative

deconvolution and Wiener filtering is influenced by the estimation accuracy of the broadening function. In addition, Wiener filtering also requires information on noise, such as the signal-to-noise ratio and power spectrum. On the other hand, the curve fitting method is more effective and is therefore the most commonly used approach in resolving overlapped peaks.

Many researchers have focused on improving the accuracy of curve fitting. For example, Zhang *et al.* [12] proposed an initial estimation method in which appropriate initial values were obtained by applying a fractional differential theory. Tan *et al.* [13] proposed an error compensation method to achieve resolution and correction of overlapping peaks. Using an iterative procedure, the fitting residual was fed back to the overlapping peaks, and multiple curve fitting processes were performed to obtain a solution with a lower residual. Furthermore, an automatic modeling algorithm was also investigated for spectra data processing. Alsmeyer and Marquardt [14] proposed an iterative

automatic modeling algorithm. The shortcoming of these methods was the requirement of a large amount of calculation or iteration.

In this work, an automatic curve fitting method is proposed to resolve the strongly overlapped peaks that could not be resolved by a spectrometer and to achieve a better resolution for the spectrometer-measured LIBS spectral data. In this proposed method, the second derivative of the LIBS spectrum is calculated by a continuous-wavelet transform (CWT). The number and positions of individual peaks of the LIBS spectrum are obtained from the local minimum of the second derivative. The full width at half-maximums (FWHMs) of individual peaks are estimated from the separation of the two maxima siding the local minimum. In addition, a threshold is introduced to eliminate small peaks in the LIBS data and to adaptively extract the major peaks with different spectral intensities. Also in the proposed method, a Trust-Region algorithm is employed to minimize the deviation between the mathematical model and the experimental LIBS spectrum. The effectiveness of the proposed method is demonstrated with both simulated and experimental LIBS spectra with overlapped peaks.

2. THEORY

Usually, in a curve fitting method to resolve overlapped peaks in LIBS spectra, there are three key factors: (1) an appropriate mathematical model to correctly describe the overlapped peaks in LIBS spectra. The line shapes of the LIBS peaks depend heavily on the line-broadening mechanisms in laser-induced plasma. The Lorentzian shape is commonly used when Stark broadening is dominant, while the Voigt shape is more accurate but requires additional fitting parameters; (2) an appropriate optimization method. The selected optimization method should have good convergence and global searching capabilities; (3) reasonable initial values for fitting parameters in the multi-parameter optimization. Appropriate initial fitting values can contribute to a faster convergence speed, a better fitting precision, a higher computation efficiency, and a shorter running time.

A. Mathematical Model

LIBS spectral lines usually show a Voigt profile due to various line-broadening mechanisms in laser-induced plasma [15]. When the noise is filtered and the baseline is corrected in LIBS data, the LIBS spectrum represents the sum of emission lines and is expressed as follows:

$$I_{th}(\lambda) = \sum_{i=1}^M I_i \cdot V(\lambda; \lambda_{0,i}, \gamma_i, \beta_i) + I_b, \quad (1)$$

with

$$V(\lambda; \lambda_0, \gamma, \beta) = \beta \cdot \sqrt{\frac{4 \ln 2}{\pi}} \cdot \frac{1}{\gamma} \cdot \exp \left[-4 \ln 2 \left(\frac{\lambda - \lambda_0}{\gamma} \right)^2 \right] + (1 - \beta) \cdot \frac{2}{\pi} \cdot \frac{\gamma}{4(\lambda - \lambda_0)^2 + \gamma^2}, \quad (2)$$

where $V(\lambda; \lambda_{0,i}, \gamma_i, \beta_i)$ represents an individual Voigt peak with wavelength λ and is described by a pseudo-Voigt function [16] shown in Eq. (2). I_i , $\lambda_{0,i}$, and γ_i are the peak height, peak

position, and FWHM, respectively. β_i is a fractional parameter satisfying $0 \leq \beta_i \leq 1$. M and i are the number of the individual peaks and the serial number of the individual peaks, respectively. I_b is a constant background representing a residual background signal due to incomplete background correction. The parameters describing the individual peaks are simultaneously determined by minimizing a square variance between the experimentally measured LIBS spectrum and the model calculation

$$\min_{I_0, \lambda_0^i, \gamma_i, \beta_i, I_b} \left\| I_m(\lambda) - \sum_{i=1}^M I_0^i \cdot V(\lambda; \lambda_{0,i}, \gamma_i, \beta_i) - I_b \right\|, \quad (3)$$

where $I_m(\lambda)$ is the measured LIBS spectrum.

Before multi-parameter minimization, the number of individual peaks in the LIBS spectrum has to be determined first. In addition, as the accuracy of the minimization depends largely on the selection of initial parameters, a reasonable estimation of the initial values is essential to the accurate extraction of the LIBS spectrum. The initial values include the number, positions, heights, and FWHMs of the individual peaks.

B. Performance of Multi-Parameter Optimization

An adjusted R-squared and root mean squared error (RMSE) are chosen to evaluate the performance of the multi-parameter minimization, that is,

$$RMSE = \left\{ \frac{1}{N} \sum_{j=1}^N [I_m(\lambda_j) - I_{th}(\lambda_j)]^2 \right\}^{1/2}, \quad (4)$$

$$R_{adj} = 1 - \left(\frac{N-1}{N-p} \right) \frac{SSE}{SST}, \quad (5)$$

where $I_m(\lambda_j)$ is the measured spectrum, and $I_{th}(\lambda_j)$ is the model spectrum calculated with Eq. (1). N is the point number of spectrum, and p is the number of regression coefficients. SSE is the sum of squared error, and SST is the sum of squared total.

C. Number and Positions of Individual Peaks

For a LIBS spectrum, the second derivative of the spectral function is usually used to determine the number and position of individual peaks, due to the fact that each peak produces a local minimum in the corresponding second derivative. In this paper, a CWT is used to calculate the second derivative and to determine the number and position of individual peaks. Mathematically, the CWT in $L^2(R)$ can be represented as [17]

$$W_f(a, b) = f * \psi_{a,b} = a^{-1/2} \int_R f(x) \psi \left(\frac{x-a}{b} \right) dx, \quad (6)$$

$$a \in R^+ - \{0\}, \quad b \in R,$$

where $f(x)$ is the signal, a is a scale, b is a translation, $\psi(x)$ is a mother wavelet, $\psi_{a,b}(x) = |a|^{-1/2} \psi(\frac{x-b}{a})$ is the scaled and translated wavelet, and W_f is a wavelet coefficient. In this work, the Mexican Hat wavelet $\psi(x) = \frac{2}{\sqrt{3}} \pi^{-1/4} (1-x^2) e^{-x^2/2}$ is selected as the mother wavelet [18]. The CWT of signal $f(x)$ is described as

$$\begin{aligned}
 W_f(a, b) &= f(x) * \psi_{a,b}(x) \\
 &= \frac{2}{\sqrt{3}} \pi^{-1/4} a^{-1/2} \int_R f(x) \left(1 - \left(\frac{x-b}{a} \right)^2 \right) e^{-\frac{1}{2} \left(\frac{x-b}{a} \right)^2} dx \\
 &= -\frac{2}{\sqrt{3}} \pi^{-1/4} a^{3/2} \cdot \frac{d^2}{db^2} \left\{ f(x) * \exp \left[-\frac{1}{2} \left(\frac{x}{a} \right)^2 \right] \right\}.
 \end{aligned} \quad (7)$$

$W_f(a, b)$ is proportional to a Gaussian smoothed second derivative of the signal. Therefore, CWT can be used to approximately calculate the second derivative, and it can effectively suppress the noise interference and improve the accuracy of peak detection.

As an example, the CWT of a Voigt peak is shown in Fig. 1. A peak produces a local minimum in the second derivative, so the local minimum can be used to determine the position of this individual peak. However, as shown in Fig. 2, artifacts may arise in the wavelet transform calculation when the wavelet scale approaches the measurement interval or the wavelet position approaches the edges of the data set. Gregoire *et al.* [19] proposed a method to circumvent this issue by modifying the discrete wavelet in the direct transform calculation

$$\Psi_{a,b}(x) = \begin{cases} \alpha \psi_{a,b}(x), & \text{if } \psi_{a,b}(x) \geq 0 \\ \beta \psi_{a,b}(x), & \text{if } \psi_{a,b}(x) < 0 \end{cases} \quad (8)$$

where α and β are chosen such that $\Psi_{a,b}(x)$ has no zeroth and first moments and has wavelet energy equal to that of the

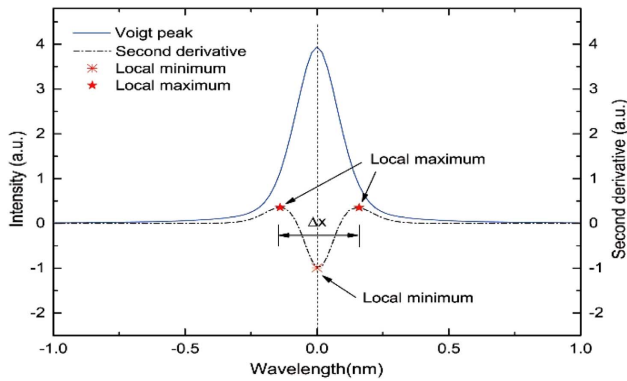


Fig. 1. Voigt peak and corresponding CWT results.

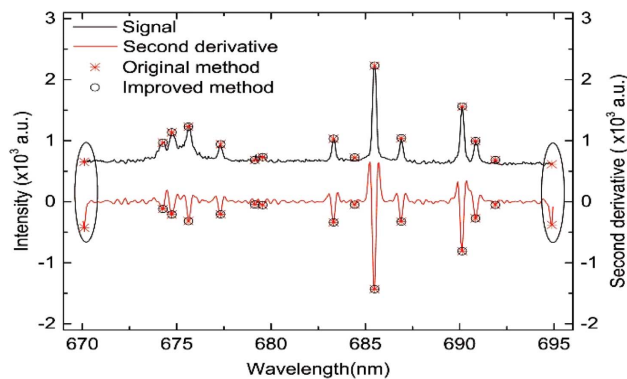


Fig. 2. Comparison of peak detection results between the original and improved algorithms.

mother wavelet $\sum_X \Psi_{a,b}(x) \delta x = \int \psi_{a,b}(x) dx$. However, this method brings complexity for programming, even though it has good performance.

In this work, a simple method to avoid the interference of artifacts is proposed. As shown in Fig. 1, for a real peak there is always one maximum on each side of the minimum, and the interference of artifacts can therefore be eliminated by checking the two maxima on both sides of the minimum. Due to artifacts only occurring at the edges of the data set, only the first and last paired maximum/minimum need to be checked. The first minimum must be excluded if the first maximum is located on the right of the first minimum. Similarly, the last minimum must be excluded if the last maximum is located on the left of the last minimum. As shown in Fig. 2, the proposed algorithm can effectively avoid the interference of artifacts and improve peak detection accuracy.

D. FWHMs and Heights of Individual Peaks

To effectively determine FWHMs of the individual peaks, the relationship between FWHM and Δx (as shown in Fig. 1, Δx is defined as the separation of two maxima on both sides of the minimum) is analyzed. As shown in Fig. 3, an approximately linear relationship exists between Δx and FWHM, which depends slightly on the wavelet scale value. FWHM can therefore be estimated roughly from the fitted linear relationship presented in Fig. 3, which is obtained by a least square method. In this work, the FWHM is estimated to be $\sim 0.7 \Delta x$.

A simple approach to determine an initial height value $I_i^{(0)}$ for an individual peak is described as follows: find the intensity $I_m(\lambda_{0,i})$ at peak position $\lambda_{0,i}$ and take the minimum of the spectrum as the background I_b . Then a reasonable initial value is the net intensity $I_i^{(0)} = I_m(\lambda_{0,i}) - I_b$.

E. Optimization Algorithms

The key steps of the proposed automatic curve fitting method are described as follows:

1. Select an appropriate wavelet scale according to the signal-to-noise ratio (SNR) and perform CWT on the original signal to obtain the second derivative. A smaller scale can obtain a higher resolution for peak detection, but it is more sensitive to noise. A larger scale can achieve a better noise suppression effect, but the resolution for peak detection will be reduced. Determine the number and positions of individual peaks from

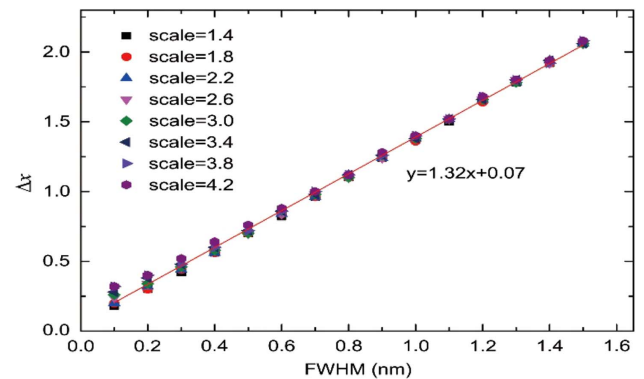


Fig. 3. Relationship between Δx and FWHM with different wavelet scales.

the local minimum of the second derivative. Eliminate the interference of artifacts by the above-mentioned method.

2. Set a threshold and extract only individual peaks in which the corresponding minimum values are greater than that threshold. This threshold is introduced for adaptively extracting the major peaks with different spectral intensities in the LIBS spectrum. Those peaks that satisfy the condition $|y| \geq \xi \cdot |y|_{\max}$ will be extracted, where ξ is the threshold, and $|y|$ is the absolute value of the local minimum of the second derivative.

3. Determine the FWHMs of individual peaks from the separation of the two maxima on both sides of the minimum. As FWHMs are strongly influenced by Stark broadening, which depends on energy level, particle species, and ionization state, all FWHMs are fitted independently in our algorithm.

4. Determine the heights of the individual peaks.

5. Minimize Eq. (3) using the Trust-Region algorithm to determine simultaneously the parameters of the individual peaks.

6. Calculate the LIBS spectrum and compare to the measurements.

In the proposed curve fitting method, all calculations are performed with a program written in MATLAB. A threshold of 0.01 can satisfy most cases. The scale a is related to the instrumental resolution; after the instrument is selected, the scale a can be pre-set and will not change in most cases. On the other hand, parameters such as number, positions, FWHMs, and heights of the individual peaks can be estimated automatically by the proposed method. From this sense, the proposed curve fitting method can be realized automatically to analyze the LIBS spectrum.

3. SIMULATION AND EXPERIMENT

Several LIBS spectra with overlapped peaks of different parameters (separation degree, peak height, and FWHM) are simulated and used to verify the feasibility of the proposed curve fitting method for resolving overlapped peaks. Each simulated spectrum is composed of two noise-free Voigt peaks. The separation degree is defined by $\kappa = |\lambda_{0,1} - \lambda_{0,2}|/[2(\gamma_1 + \gamma_2)]$, and λ_0 and γ are the peak position and FWHM of individual peaks, respectively [12]. A wavelength interval of 0.05 nm is chosen according to the spectral resolution of the spectrometer used in our experiment.

In addition, the proposed curve fitting method is also applied to resolve individual peaks in experimental LIBS spectra. The LIBS experimental setup is described as follows. A Q-switched 1064 nm Nd:YAG laser (Continuum Surelite, with maximum pulse energy 700 mJ, pulse width 18 ns, and repetition rate 1–10 Hz) is focused into gas samples (compositions of N_2/SF_6 gas mixtures are listed in Table 1) by a focusing lens with a 50 mm focal length. The plasma emission is detected and recorded by a HORIBA iHR320 JY spectrometer (equipped with a 1200 grooves/mm grating and the corresponding wavelength interval is 0.052 nm) equipped with an intensified charge-coupled device (ICCD). For LIBS spectrum measurements, the laser pulse energy is 50 mJ and the repetition rate is 5 Hz. The slit width of the spectrometer is set to 50 μ m. The wavelength of the spectrometer is calibrated by a

Table 1. Components of N_2/SF_6 Gasses Mixture Used in the Experiment

Element	Concentration (%)				
	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
F	0.8333	0.8108	0.7692	0.7317	0.6977
S	0.1389	0.1351	0.1282	0.1220	0.1163
N	0.0278	0.0541	0.1026	0.1463	0.1861

mercury lamp, and its spectral response is calibrated by a tungsten lamp and a deuterium lamp. Instrumental line-broadening is measured by a single-frequency He–Ne laser (with center wavelength 632.8 nm and FWHM 0.29 pm). With 50 μ m slit width, the instrumental broadening shows a Lorentzian profile with a FWHM of 0.1367 nm, and its effect on the measured LIBS spectrum is considered in the data processing.

4. RESULTS AND DISCUSSION

The simulated LIBS spectra with overlapped peaks of different parameters and corresponding resolving results with the proposed curve fitting method are shown in Fig. 4. The scale of CWT is chosen to be $a = 0.48$, the FWHMs of the individual peaks are chosen as $\gamma_1 = 0.18$ nm and $\gamma_2 = 0.22$ nm in Figs. 4(a) and 4(b), $\gamma_1 = 0.18$ nm and $\gamma_2 = 0.36$ nm in Figs. 4(c) and 4(d), respectively. The results show that the proposed method can accurately resolve the overlapped peaks, even with a small separation degree, but the minimum resolvable separation degree κ_m depends on the FWHM ratio γ_1/γ_2 and strength ratio I_1/I_2 of the two individual peaks.

Figure 5 shows how spectral parameters such as strength ratio I_1/I_2 , FWHM ratio γ_1/γ_2 , and wavelet scale a influence the resolving of individual peaks with the proposed method. As shown in Fig. 5(a), the peak height (intensity) ratio I_1/I_2 also affect κ_m . κ_m increases with the increasing difference of peak heights. Figure 5(b) shows the relationship between κ_m and γ_1/γ_2 . As FWHM ratio increases, κ_m first increases, reaches a maximum, and then decreases. Therefore, a larger difference between γ_1 and γ_2 results in a smaller κ_m . Figure 5(c) shows the relationship between κ_m and the scale a . The result indicates that a smaller scale a contributes to a better resolving performance of individual peaks. From the results presented in Fig. 5, the minimum resolvable separation degree is dependent largely on the strength ratio, the FWHM ratio, as well as the wavelet scale. In the case that the spectrometer resolution is 0.05 nm ($a = 0.48$) and the FWHMs of the two individual peaks are comparable, the lowest minimum resolvable separation degree is approximately 0.27 [as shown in Fig. 5(a)].

Experimentally, as an example, Fig. 6(a) shows the measured LIBS spectrum of 37.5% N_2 + 62.5% SF_6 gas mixture in the wavelength range of 400–850 nm. The proposed method is applied to resolve individual peaks in two overlapping regions as marked by red boxes in Fig. 6(a). The corresponding curve fitting results are shown in Figs. 6(b) and 6(c), respectively. The measured spectra, the corresponding curve fitting results, the individual peak profiles, and the corresponding positions (which were obtained by CWT with scale $a = 0.5$ and

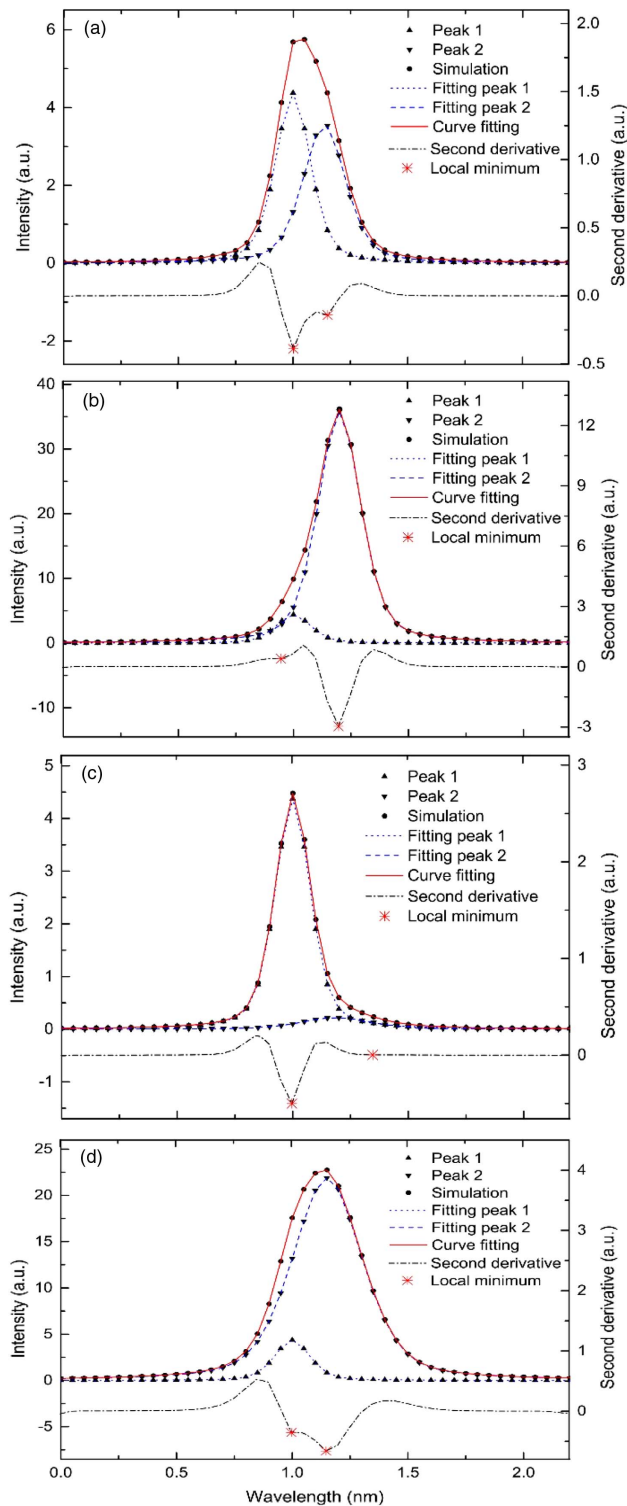


Fig. 4. Simulated LIBS spectra with overlapped peaks of different parameters and corresponding results resolving with the proposed method. (a) $\kappa = 0.34$, $I_1/I_2 = 1$; (b) $\kappa = 0.50$, $I_1/I_2 = 0.1$; (c) $\kappa = 0.34$, $I_1/I_2 = 10$; and (d) $\kappa = 0.28$, $I_1/I_2 = 0.1$.

$\xi = 0.01$) are presented in the upper diagram, and the model residuals are shown in the lower diagram in Figs. 6(b) and 6(c), respectively. The adjusted R-squared values are 0.9974 in Fig. 6(b) and 0.9980 in Fig. 6(c). All experimental data are

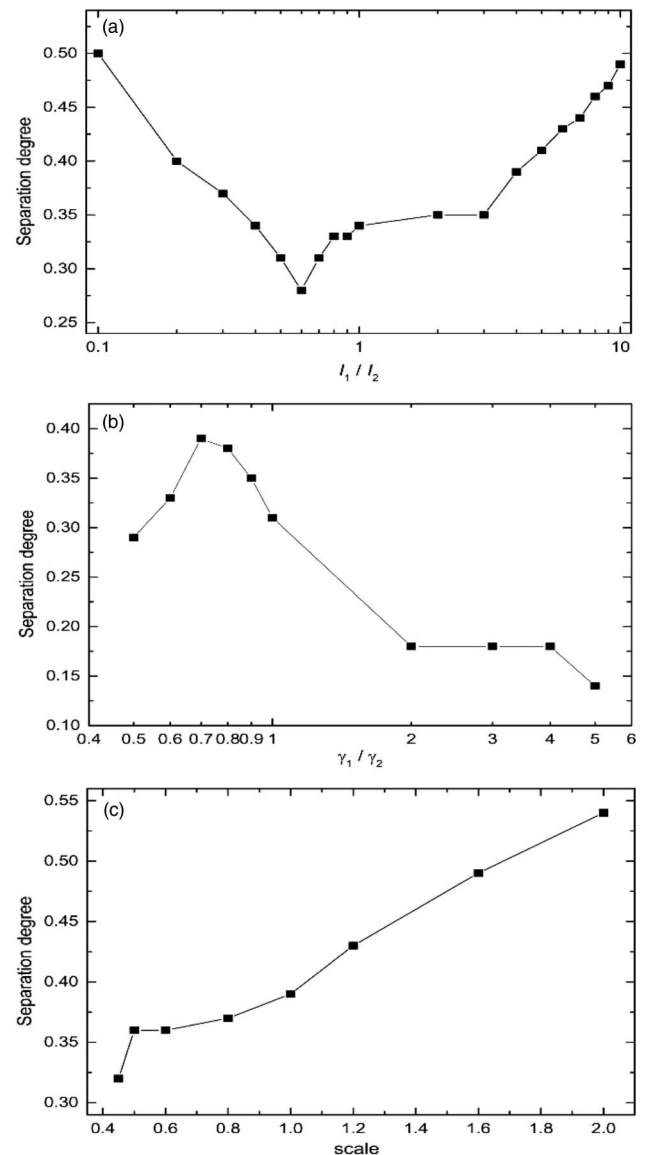


Fig. 5. Relationship between (a) the minimum resolvable separation degree κ_m and strength ratio I_1/I_2 , calculated with scale of CWT $a = 0.48$, FWHMs of individual peaks $\gamma_1 = 0.18$ nm and $\gamma_2 = 0.22$ nm. (b) The minimum resolvable separation degree κ_m and FWHM ratio γ_1/γ_2 , calculated with $a = 0.48$, strength ratio of individual peaks $I_1/I_2 = 1$. (c) The minimum resolvable separation degree κ_m and wavelet scale a , calculated with $\gamma_1 = 0.18$ nm and $\gamma_2 = 0.22$ nm, strength ratio of individual peaks $I_1/I_2 = 1$.

ASCII format, and the sizes are 7890, 195, and 105 points for a wavelength range of 400–850 nm, 816–826 nm, and 729–735 nm, respectively. The algorithm running times are 2.4 s and 2.3 s for 816–826 nm and 729–735 nm, respectively. The results clearly indicate that the experimental LIBS spectrum with severely overlapped peaks can be accurately resolved by the proposed curve fitting method. It is noted that the relationship of $\text{FWHM} \sim 0.7\Delta x$ may deviate as the overlapping degree increases (for example, $\text{FWHM} \sim 0.6\Delta x$ or $0.5\Delta x$ might become more reasonable). Still, the estimated FWHMs are reasonable, and the fitting results converge. However, the

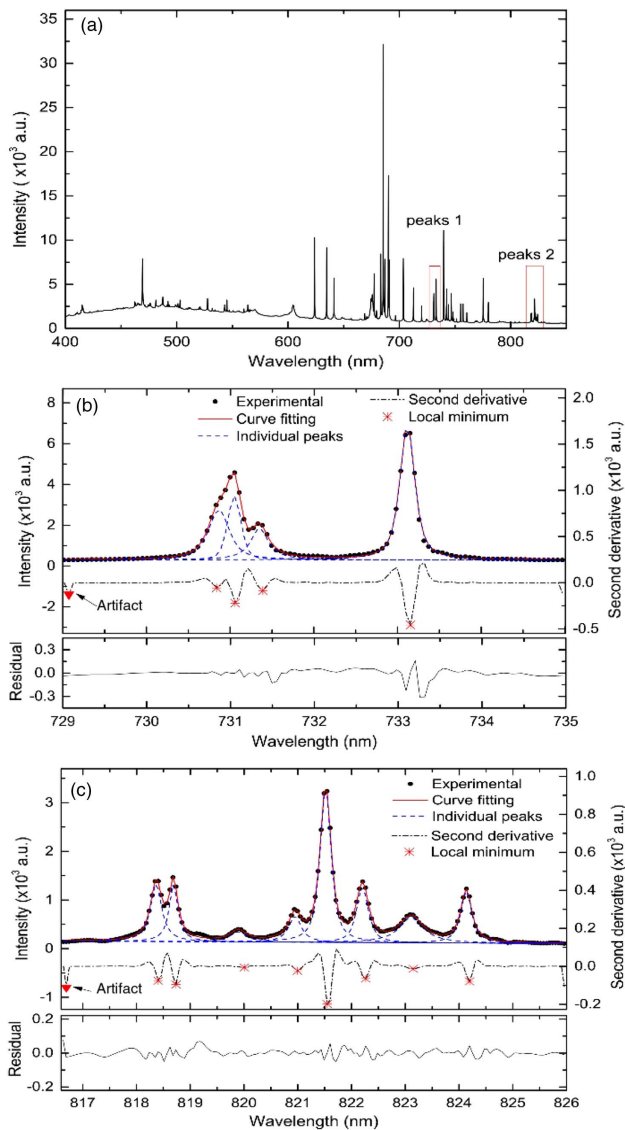


Fig. 6. (a) LIBS spectrum of N_2/SF_6 gas mixture and resolved overlapping peaks in different wavelength regions (b) 729–735 nm and (c) 816–826 nm.

algorithm might fail if the SNRs of the LIBS spectrum data are too low.

To verify the effectiveness of the proposed curve fitting method in LIBS data processing for composition analysis application, a calibration experiment with N_2/SF_6 gas mixtures of different concentration ratios listed in Table 1 is performed. The calibration results before and after resolving the overlapped peaks with the proposed method are shown in Fig. 7. In Fig. 7(a), calibration curves are obtained between the concentration and intensity of the F 730.901 nm line [the first individual peak shown in Fig. 6(b)] in the LIBS spectrum. The fitted intensity represents the peak height after resolving, which is obtained from Eq. (1), while peak intensity represents the measured peak intensity, with the peak position obtained via the proposed method. Comparing the calibration results before and after resolving the overlapped peaks with the proposed method, the linear fit between the F 730.901 nm line intensity

and F concentration is improved from $R^2 = 0.8731$ to 0.9972 by effectively resolving the overlapped peaks, even with a low separation degree (~ 0.40 , calculated value), with the proposed curve fitting method. Furthermore, N 818.4861 nm,

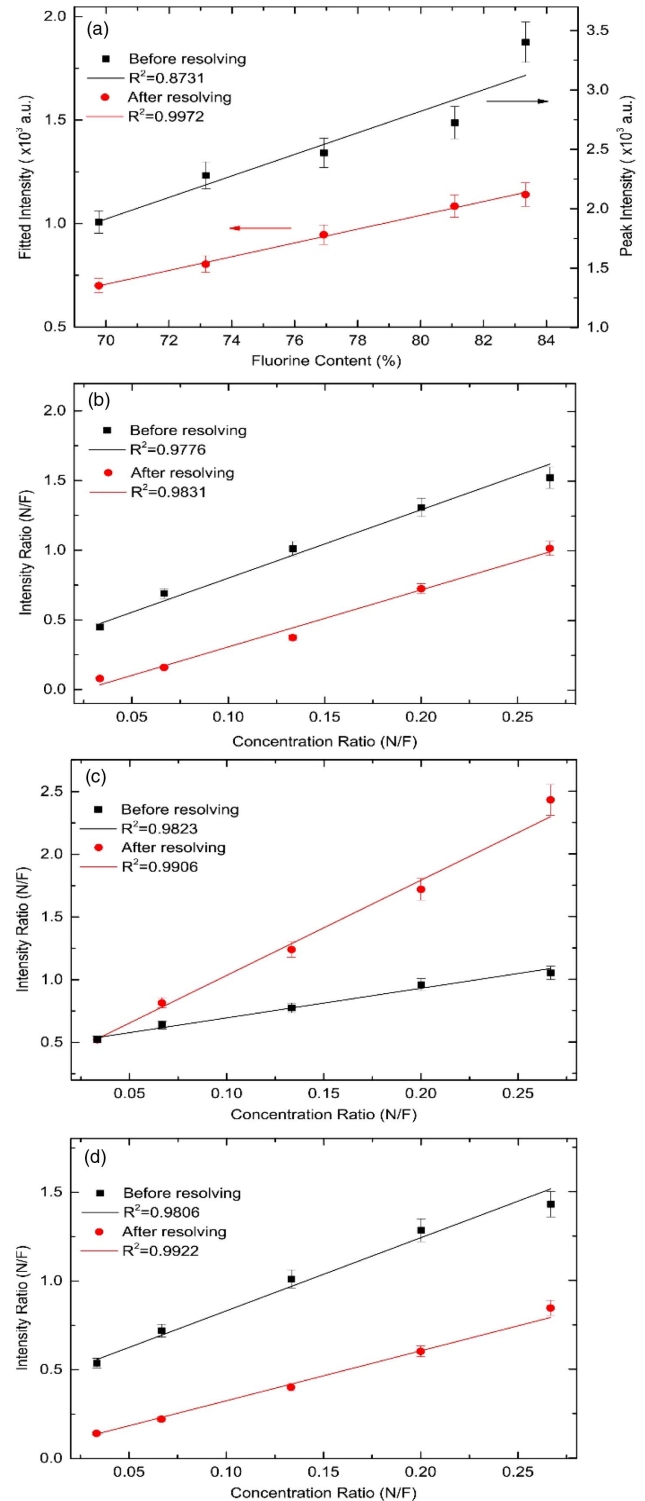


Fig. 7. (a) Calibration curves between the F concentration and intensity of the F 730.901 nm line; (b)–(d) the calibration results between the N/F concentration ratio and intensity ratio of (b) N 818.4861 nm, (c) N 821.0715 nm, and (d) N 822.3128 nm to the F 823.077 nm line.

N 821.0715 nm, N 822.3128 nm, and F 823.077 nm lines [the first, fourth, sixth, and seventh individual peaks in Fig. 6(c)] can also be selected to establish calibration curves for N/F composition analysis, as shown in Figs. 7(b)–7(d). The calibration curves are established between the N/F concentration ratio and the intensity ratio of the corresponding nitrogen line to the fluorine F 823.077 nm line. The F 823.077 nm line is chosen as the internal standard line. After resolving the overlapped peaks with the proposed method, the resulting adjusted R-squared values improve from 0.9776 to 0.9831, 0.9823 to 0.9906, and 0.9806 to 0.9902, respectively, indicating better linear fits between the N/F concentration ratio and the intensity ratio of nitrogen to fluorine lines. Furthermore, the performance of the proposed method further improves with the increasing overlapped degree. These results demonstrate that the proposed curve fitting method is effective in accurately extracting the line parameters of individual peaks in LIBS data for quantitative composition analysis.

5. CONCLUSIONS

In conclusion, a CWT-based automatic curve fitting method was proposed for resolving severely overlapped peaks in the LIBS spectrum. CWT was employed to calculate the second derivative of the LIBS spectrum, from which the number, positions, and FWHM linewidths of individual peaks were determined and used as initial parameter values in multi-parameter optimization to accurately extract the spectral parameters of major individual peaks in the LIBS data. Experimentally, for LIBS analysis of a N_2/SF_6 gas mixture, the extracted spectral parameters from severely overlapping peaks were used to establish a calibration curve between F concentration and a LIBS intensity of F 730.901 nm line with a better linear fit (with adjusted R-squared value improved from 0.8731 to 0.9972), as well as between the N/F concentration ratio and intensity ratio of the corresponding nitrogen line to fluorine F 823.077 nm line (with adjusted R-squared value improved to be >0.98 and 0.99) for quantitative composition analysis. Furthermore, even better performance is expected with more seriously overlapped peaks. The proposed automatic curve fitting method has great potential for online and real-time application of LIBS, as well as for other spectral signal processing.

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